ECEN-629: Homework 1

Due on September 16, 2025 at 12:45 pm

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Problem 1

Compute the norm of the following vector $x = (-100, -99, -98, \dots, 99, 100)^T$.

(a) ℓ_1 norm;
(b) ℓ_∞ norm;
(c) ℓ_2 norm.

Solution

We have the definition of a norm as

$$\|x\|_p = \left(\sum_{i=1}^{n} |x_i|^p\right)^{1/p}$$

Then for each part we have:

$$\begin{aligned}
\ell_1 : \|x\|_1 &= |-100| + |-99| + \dots + |99| + |100| \\
&= 2(1 + 2 + \dots + 99 + 100) \\
&= 2 \cdot \frac{100 \cdot 101}{2} \\
&= 10100
\end{aligned}$$

$$\begin{aligned}
\ell_\infty : \|x\|_\infty &= \max\{x\} \\
&= 100
\end{aligned}$$

$$\begin{aligned}
\ell_2 : \|x\|_2 &= \sqrt{(-100)^2 + (-99)^2 + \dots + (99)^2 + (100)^2} \\
&= \sqrt{2(1^2 + 2^2 + \dots + 99^2 + 100^2)} \\
&= \sqrt{2 \cdot \frac{100 \cdot 101 \cdot 201}{6}} \\
&= \sqrt{676700} \\
&\approx 822.62
\end{aligned}$$

Problem 2

For a quadratic function $f(x) = x^T Px + q^T x + r$ where P is not necessarily symmetric, show that the gradient is $\nabla f(x) = (P + P^T)x + q$. Find the Hessian of the function. Provide details in your proofs.

Solution

I start be decomposing $f(x)$ into summations:

$$f(x) = \sum_{i=1}^{m} \sum_{j=1}^{m} P_{ij} x_i x_j + \sum_{i=1}^{m} q_i x_i + r$$

Then we find the partials of the first term:

$$\frac{\partial}{\partial x_k} f(x) = \frac{\partial}{\partial x_k} \left(\sum_{i=1}^{m} \sum_{j=1}^{m} P_{ij} x_i x_j \right) = \sum_{i=1}^{m} \sum_{j=1}^{m} \frac{\partial}{\partial x_k} (P_{ij} x_i x_j)$$
$$= \sum_{i=1}^{m} P_{kj} x_j + \sum_{j=1}^{m} P_{ik} x_i$$

And the second term:

$$\frac{\partial}{\partial x_k} \left(\sum_{i=1}^{m} q_i x_i + r \right) = q_k$$

The final term r is constant, so its derivative is zero.

Now we can combine these indexed derivatives to get the gradient:

$$\nabla f(x) = (P + P^T)x + q$$

Now the Hessian of $f(x)$ is found by taking the partials of the gradient we just found:

$$\mathbf{H} = \nabla^2 f(x) = P + P^T$$

Problem 3

A matrix $A \in \mathbb{R}^{m \times n}$ can also be viewed as a vector, by stacking all its n columns into a single length-mn column. Let $x \in \mathbb{R}^m$, $y \in \mathbb{R}^n$ be two column vectors. Show that

$$\mathrm{tr}(A^T x y^T) = \left\langle x y^T, A \right\rangle = x^T A y,$$

where tr is the trace of the matrix.

Solution

We know that rotation of our trace elements gives us

$$\mathrm{tr}(A^T x y^T) = \mathrm{tr}(y^T A^T x)$$

Here the product inside the trace is a 1×1 matrix, thus

$$\mathrm{tr}(y^T A^T x) = y^T A^T x = x^T A y$$

The inner product is defined as

$$\left\langle x y^T, A \right\rangle = \sum_{i=1}^{m} \sum_{j=1}^{n} (x y^T)_{ij} A_{ij} = \sum_{i=1}^{m} \sum_{j=1}^{n} x_i y_j A_{ij} = x^T A y$$

Thus both the trace of $A^T x y^T$ and the inner product $\left\langle x y^T, A \right\rangle$ equal $x^T A y$.

Problem 4

Let $B \in \mathbb{R}^{n\times n}$ be a given symmetric matrix of size n-by-n. Determine if the following set is convex: the set of all n-by-n symmetric A matrices, such that $A - B$ is positive semidefinite. If it is convex, provide your reasoning; if not, provide a counter-example.

Solution

Putting it in set notation, we have

$$C = \{A \in \mathbb{S}^n \mid A - B \succeq 0\}$$

Now we know that a set is convex if

$$A_1, A_2 \in C, \quad 0 \leq \theta \leq 1 \quad \implies \quad \theta A_1 + (1-\theta)A_2 \in C$$

Writing out the set with some $A_1, A_2 \in C$, we have

$$\theta A_1 + (1-\theta)A_2 - B \succeq 0$$

Distributing the subtraction of B, we have

$$\theta(A_1 - B) + (1-\theta)(A_2 - B) \succeq 0$$

Since $A_1, A_2 \in C$, we know that $A_1 - B \succeq 0$ and $A_2 - B \succeq 0$. Since this is a linear combination of PSD's, it is also PSD. We have now seen that $\theta A_1 + (1-\theta)A_2 \in C$, so the set is convex.

Problem 5

A set is defined as

$$C = \{x \in \mathbb{R}^n \mid \|x - a\|_2 \le \theta \|x - y\|_2 \text{ for all } y \in S\}$$

where $S \subseteq \mathbb{R}^n$.

(a) Determine whether the set is convex, which may depend on the value of θ. For the case that it is, provide your reasoning; otherwise, provide a counter-example.

(b) Let $a = (1, 0)^T$ and $S = \{(-1, 0)^T, (0, -1)^T\}$. For $\theta = 2$ and $\theta = 0.5$, sketch the resultant set, respectively.

Solution

Part (a): I will define two arbitray points in C, now $x_1, x_2 \in C$ and check for convexity. I'll use $\phi x_1 + (1 - \phi)x_2$ to avoid confusion with θ.

$$\|\phi x_1 + (1 - \phi)x_2 - a\|_2 \le \theta \|\phi x_1 + (1 - \phi)x_2 - y\|_2, \quad \forall y \in S$$

Which equals

$$\|\phi(x_1 - a) + (1 - \phi)(x_2 - a)\|_2 \le \theta \|\phi(x_1 - y) + (1 - \phi)(x_2 - y)\|_2, \quad \forall y \in S$$

The value of θ determines the size and shape of the set.

Part (b): For $\theta = 0.5$, the set is defined as s

$$\{x \in \mathbb{R}^n \mid \|x - (1, 0)^T\|_2 \le 0.5 \|x - y\|_2 \text{ for all } y \in S\}$$

Which simplifies to the regions $(x_1 - 1)^2 + x_2^2 \le 0.25((x_1 + 1)^2 + x_2^2)$ and $(x_1 - 1)^2 + x_2^2 \le 0.25(x_1^2 + (x_2 + 1)^2)$. Plotting those regions

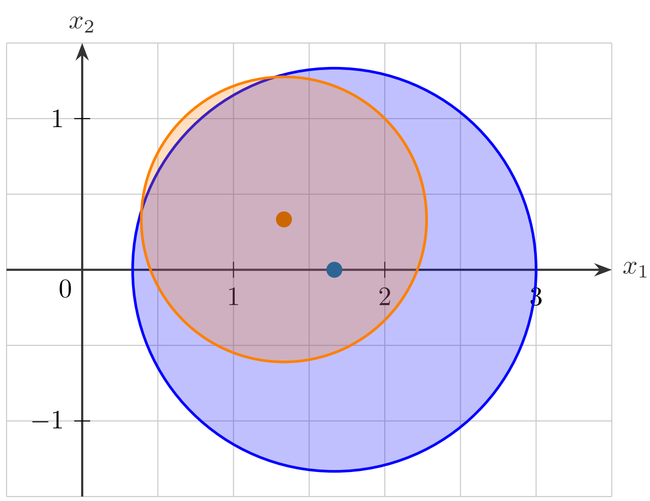


Figure 1: Set for $\theta = 0.5$

Problem 5 continued on next page...

For $\theta = 2$, the set is defined as

$$\{x \in \mathbb{R}^n \mid \left\|x - (1,0)^T\right\|_2 \leq 2 \left\|x - y\right\|_2 \text{ for all } y \in S\}$$

Which simplifies to the regions $(x_1-1)^2+x_2^2 \leq 4((x_1+1)^2+x_2^2)$ and $(x_1-1)^2+x_2^2 \leq 4(x_1^2+(x_2+1)^2)$.

Plotting those regions

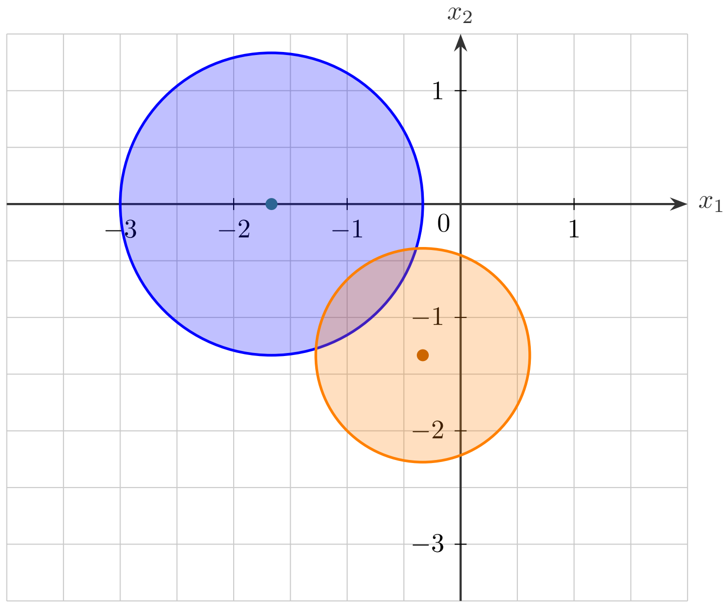


Figure 2: Set for $\theta = 2$

Problem 6

(a) Show that if S_1 and S_2 are convex set in $\mathbb{R}^n$, then the set sum

$$S \triangleq \{x_1 + x_2 \mid x_1 \in S_1,\, x_2 \in S_2\}$$

is also convex.

(b) Let $S_1 = \{(-1,-1)^T\}$ and $S_2 = \{(2,2)^T + u \mid \|u\| \leq 1\}$. What is their set sum?

(c) Show that if S_1 and S_2 are convex set in $\mathbb{R}^n$, then the inverse sum

$$S \triangleq \bigcup_{1 \geq \lambda \geq 0} \{\lambda S_1 \cap (1-\lambda) S_2\}$$

is also convex, where $\lambda S_j = \{\lambda x_j \mid x_j \in S_j\}$, $j = 1, 2$.

(d) Let $S_1 = \{(1,1)^T\}$ and $S_2 = \{(2,2)^T + u \mid \|u\| \leq 1\}$. What is their inverse sum?

Solution

Part (a): Similar to above, we choose two arbitray points in S

$$a = x_1 + x_2, \quad b = y_1 + y_2, \quad x_1, y_1 \in S_1, \quad x_2, y_2 \in S_2$$

and check if they satisfy the conditions of a convex set

$$\begin{aligned} \theta a + (1-\theta) b &= \theta(x_1 + x_2) + (1-\theta)(y_1 + y_2) \\ &= \underbrace{\theta x_1 + (1-\theta) y_1}_{\in S_1} + \underbrace{\theta x_2 + (1-\theta) y_2}_{\in S_2} \end{aligned}$$

So this is a sum of terms $x_1 + x_2$ where $x_1 \in S_1$ and $x_2 \in S_2$, so $\theta a + (1-\theta) b \in S$. Thus, S is a convex set.

Part (b): The set sum of S_1 and S_2 is

$$\begin{aligned} S_1 + S_2 &= \{(-1,-1)^T + (2,2)^T + u \mid \|u\| \leq 1\} \\ &= \{(1,1)^T + u \mid \|u\| \leq 1\} \end{aligned}$$

Which is a circle of radius 1 centered at (1, 1).

Part (c): We choose two arbitray points $x, y \in S$

Part (d):

Problem 7

It is known that one halfspace contains another in the following form

$$\{x \mid a^T x \le b\} \subseteq \{x \mid c \cdot a^T x \le \tilde{b}\}.$$

Provide the conditions that the parameters c, b, and $\tilde{b}$ must satisfy. Are these conditions also sufficient for the containment relation between these two halfspaces?

Solution

For the first set to be a subset of the second, any x that satisfies $a^T x \le b$ must also satisfy $c \cdot a^T x \le \tilde{b}$.

If $c < 0$, then x must satisfy

$$\begin{cases} a^T x \le b \\ c \cdot a^T x \le \tilde{b} \end{cases} \implies \begin{cases} a^T x \le b \\ a^T x \ge \tilde{b}/c \end{cases}$$

Which presents a contradiction, so $c \ge 0$.

For $c = 0$, we have

$$\begin{cases} a^T x \le b \\ 0 \le \tilde{b} \end{cases}$$

Which is valid when $\tilde{b} \ge 0$.

For $c > 0$

$$\begin{cases} a^T x \le b \\ c \cdot a^T x \le \tilde{b} \end{cases}$$

Which requires that $b \le \tilde{b}/c$.

Problem 8

For a pair of given sets $C, D \subseteq \mathbb{R}^n$, consider the following set

$$S = \{(a, b) \in \mathbb{R}^{n+1} \mid a^T x \leq b \text{ for all } x \in C,\, a^T x \geq b \text{ for all } x \in D\}.$$

Prove that the set S is convex.

Solution

Choosing two arbitrary points in S, we have

$$x = (a_1, b_1), \quad y = (a_2, b_2)$$

and applying it to our convex definition

$$\theta x + (1-\theta)y = (\theta a_1 + (1-\theta)a_2,\, \theta b_1 + (1-\theta)b_2)$$

Now to see if this is in S, we check the two conditions, starting with the first

$$(\theta a_1 + (1-\theta)a_2)^T x = \theta a_1^T x + (1-\theta)a_2^T x \leq \theta b_1 + (1-\theta)b_2, \quad \forall x \in C$$

and then the second

$$(\theta a_1 + (1-\theta)a_2)^T x = \theta a_1^T x + (1-\theta)a_2^T x \geq \theta b_1 + (1-\theta)b_2, \quad \forall x \in D$$

Since both of these conditions hold by the set definition, the set S is convex.

Problem 9

Let $C = \{X \in \mathbb{R}^{n\times n} \mid a^T X a \geq 0, \forall a \in \mathbb{R}^n\}$. Determine whether the set

$$\mathcal{S} = \{A \in \mathbb{R}^{n\times n} \mid A - M \in C\}$$

is convex, where $M \in C$ is a fixed matrix. Provide justification or a counterexample. Note that usually the definition of a matrix being positive-definite requires the matrix to be symmetric, but this problem removes that condition.

Solution

Again choose two arbitrary points in $\mathcal{S}$

$$A_1, A_2 \in \mathcal{S}$$

and check the convex condition

$$\theta A_1 + (1-\theta)A_2 - M \in C$$

Distributing the subtraction of M, we have

$$\theta(A_1 - M) + (1-\theta)(A_2 - M) \in C$$

Now this new matrix is in C if

$$a^T(\theta(A_1 - M) + (1-\theta)(A_2 - M))a \geq 0$$

Which becomes

$$a^T(\theta(A_1 - M))a + a^T((1-\theta)(A_2 - M))a \geq 0$$

Since $A_1, A_2 \in \mathcal{S}$, we know that $A_1 - M \in C$ and $A_2 - M \in C$, so these terms are ≥ 0. Set $\mathcal{S}$ satisfies the convex conditions, so it is convex.

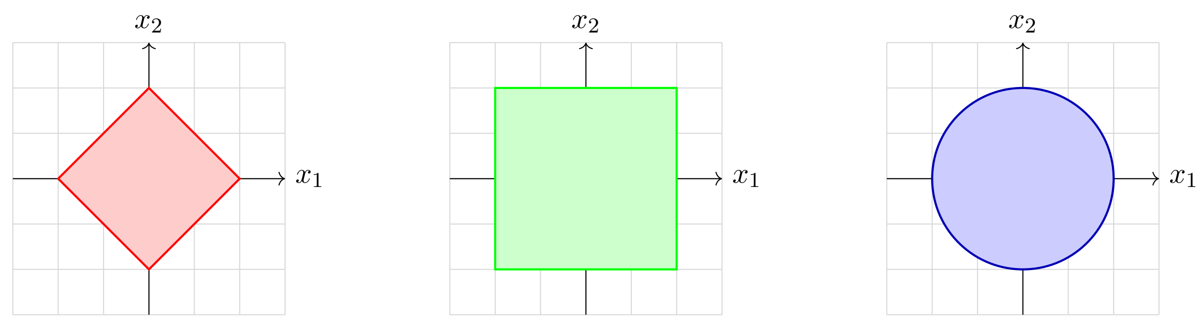
Problem 10

Plot the norm balls of radius $r = 1$ for the following norm in $\mathbb{R}^2$ and $\mathbb{R}^3$:

(a) ℓ_1 norm;
(b) ℓ_∞ norm;
(c) ℓ_2 norm.

Solution

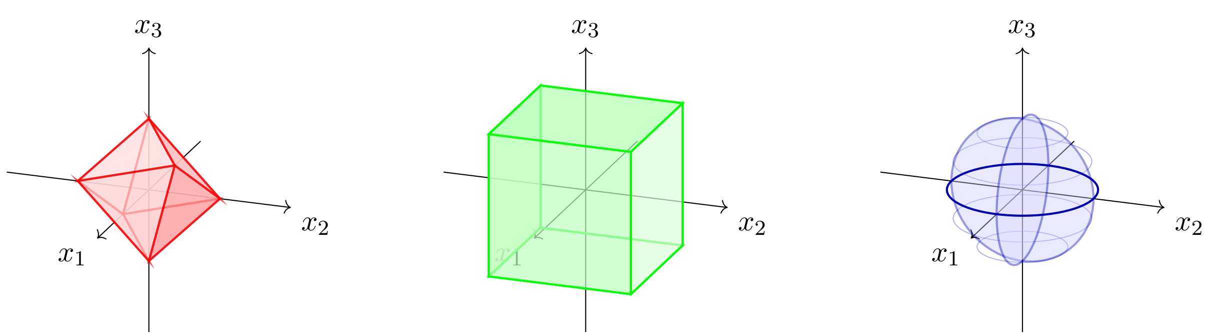
$\mathbb{R}^2$



(a) ℓ_1 norm (b) ℓ_∞ norm (c) ℓ_2 norm

Figure 3: Norm balls in $\mathbb{R}^2$

$\mathbb{R}^3$



(a) ℓ_1 norm (b) ℓ_∞ norm (c) ℓ_2 norm

Figure 4: Norm balls in $\mathbb{R}^3$